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**Subject : Thermodynamics - I / Thermodynamics/
Pr. of Therm. Engg./ Energy Dynamics / Engineering
Thermodynamics**

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Perfect vacuum and zero pressure occurs at

- a) 0°C b) 273°C
c) 273K d) -273°C

Q.2 The substance whose evaporation from liquid state is complete, is known as

- a) Vapour b) Perfect Steam
c) Real gas d) Perfect gas

Q.3 The change of entropy when a gas rejects heat is given by

- Negative value
- Positive value

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- c) May be negative or positive
 - d) Zero
- Q.4 The water at a height of hill top boils at a temperature
- a) More than 100°C
 - b) Less than 100°C
 - c) Same as 100°C
 - d) Depends on temperature at hill
- Q.5 Mollier diagram is a plot between
- a) h-s
 - b) t-s
 - c) h-t
 - d) p-v
- Q.6 Which of the following is boiler mounting?
- a) Economizer
 - b) Superheater
 - c) Pressure gauge
 - d) Air preheater

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 The total heat contained in a gas is known as _____
- Q.8 The heat required to raise the temperature of one kg of mass of a substance by one degree is known as _____

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- Q.9 Second law of thermodynamics is based on conservation of energy (True/False)
- Q.10 For dry steam dryness fraction is _____
- Q.11 In _____ boiler the water tubes are inclined to promote water circulation.
- Q.12 The intercooling in multistage compressor is done to _____

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Define enthalpy & give its expression for constant volume process.
- Q.14 Explain free expansion with example.
- Q.15 Explain the need and working of economiser with a diagram.
- Q.16 Classify compressors.
- Q.17 Derive an expression for specific heats and give the relation between them.
- Q.18 Define steady flow energy equation. Give its limitations.
- Q.19 Explain P-V-T surface of an ideal gas.
- Q.20 Define characteristic gas constant and give its derivation.

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- Q.21 One kg of a steam is at a pressure of 10 bar, with dryness fraction 0.9 Calculate its enthalpy.
- Q.22 Explain the concept of irreversibility and entropy.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the working of Lancashire boiler with the help of a neat diagram.
- Q.24 A gas occupies a volume 0.1 cubic meter at a temperature 25° C and a pressure of 1.5 bar. Find the final temperature of the gas. If it is compressed to a pressure of 7.5 bar and occupies a volume of 0.05 cubic meter.
- Q.25 Give at least eight differences between reciprocating and rotary compressors.